

PANDA Component Description: *timer_8bit*

Overview

The `timer_8bit` component implements an 8-bit timer with adjustable prescaler and user settable auto reload value. The timer can be configured through a configuration register which controls the following options: run/stop, up/down-counter, auto reload enable and interrupt enable.

The prescaler is directly calculated from the two generics. Therefore the user has only to know the clock frequency and the desired timer resolution.

Generics

Name	Type	Default	Description
<code>clk_frequency</code>	integer	10'000'000	Clock frequency in Hz
<code>resolution_us</code>	integer	1'000	Timer Resolution in μ s (1...1'000'000)

Registers

Timer Register

Name: TMR
Address: Base + 0x00

Name	TMR
R/W	R/W
Reset Value	0x00

The *TMR* register reflects the current value of the timer. It can be read and written. Every write operation to *TMR* resets the prescaler to provide exact timing generation.

Reload Register

Name: RLD
Address: Base + 0x01

Name	RLD
R/W	R/W
Reset Value	0x00

The *RLD* register holds the value, which is loaded into *TMR* if an overflow occurs and the auto reload option is enabled (*RELOAD* in *CFG* is set).

Configuration Register

Name: CFG

Address: Base + 0x02

Name	-	-	-	-	UPDOWN	RELOAD	RUN	INTON
R/W	-	-	-	-	R/W	R/W	R/W	R/W
Reset Value	0	0	0	0	0	0	0	0

INTON: '1' → State of *INT* is forwarded to the interrupt line
 '0' → No interrupts are generated

RUN: '1' → Timer runs
 '0' → Timer is stopped

RELOAD: '1' → *TMR* is loaded with the value of *RLD* after an overflow
 '0' → *TMR* is not changed after an overflow

UPDOWN: '1' → Timer runs up, overflow from 0xFF to 0x00
 '0' → Timer runs down, overflow from 0x00 to 0xFF

Interrupt Register

Name: INT

Address: Base + 0x03

Name	INT						
R/W	-	-	-	-	-	-	R/W
Reset Value	-	-	-	-	-	-	0

INT is set on every overflow of *TMR* regardless of the state of *INTON* in *CFG*. It must be cleared by Software.

Driver Macros

START

Call: *_START

Starts the timer (sets *RUN* in *CFG*).

STOP

Call: *_STOP

Stops the timer (clears *RUN* in *CFG*).

CLRINT

Call: *_CLRINT

Clears the interrupt (clears *INT*).

INIT_CONTINT

Call: *_INIT_CONTINT [intervall]

Parameter	Description	Immediate/Variable	Size
intervall	Interrupt intervall in ticks	Immediate	8-Bit

Initializes the timer for continuous interrupt generation. The general IE bit of the CPU must be set by user software.

INIT_WAITNTICKS

Call: *_INIT_WAITNTICKS

Initializes the timer for the usage of the WAITNTICKS macro.

WAITNTICKS

Call: *_WAITNTICKS [N]

Parameter	Description	Immediate/Variable	Size
N	Delay time in ticks	Immediate/Variable	8-Bit

Waits for the specified time. This macro can replace delay loops.